

UTSAVA

Complete Project Development Plan — v2

The single build reference for Utsava: scope, users and permissions, the Next.js + Supabase architecture, data model and security design, the sprint-level delivery plan with the photography-first wedge built in, team, budget, quality practices, risks and the launch gate. Consolidates and supersedes the Development Plan v1, the Next.js + Supabase Architecture note, and the build-relevant parts of the Photography-First Strategy. The Business Plan v1 remains the commercial reference.

Prepared for	Vinay
Date	July 2026
Stack	Next.js App Router (Vercel) · Supabase · Expo React Native
Fixed milestones	MVP Dec 2026 · Beta Feb 2027 · Public launch Apr 2027 (Bengaluru + Delhi NCR) · Escrow Jul 2027
Build budget	₹2.8 Cr over 18 months (product & engineering allocation of the ₹7 Cr seed)
Wedge	Photography-first category strategy; founder's studio as anchor vendor #1 from Oct 2026

“Utsava” is a working title. Dates, estimates, prices and tool choices are planning assumptions to re-validate at build time; not legal or financial advice.

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1. Overview & fixed constraints

Utsava is an all-events discovery and booking marketplace for India — weddings, social celebrations and corporate events — with photography as the wedge category. This plan turns the business commitments into a buildable programme. Four dates are immovable: **MVP feature-complete December 2026** (the field team needs product to onboard 1,000 listings before year-end), **closed beta with 200 customers February 2027**, **public launch April 2027** with 3,000 live listings (800 of them photographers), and **escrow bookings July 2027**. The budget envelope is ₹2.8 Cr over 18 months, of which roughly ₹1.5 Cr lands before launch — matching the financial model's pre-launch build line.

Three design principles govern every trade-off. **Supply tooling before demand product:** the vendor onboarding tool ships in September 2026, seven months before customers arrive, because 3,000 concierge-quality listings are the launch asset. **Lead quality is an engineering feature:** OTP verification, budget and date capture, the five-vendor cap and honest vendor dashboards are core MVP scope — renewal, the model's most sensitive assumption, depends on them.

Photography does double duty: the category converts fully online, its vendors are the proven payers, and every shoot generates the galleries and real-wedding stories that market the platform — including from the founder's own anchor studio, live from October 2026.

2. Scope — MoSCoW tiers

Tier	Scope	Notes
Must — launch (Apr 2027)	Category × locality discovery with price bands and the 'free on my date' availability filter; portfolio-first vendor profiles with style taxonomy (candid/traditional/cinematic...); package cards normalised to per-day pricing; shortlists; enquiry flow with OTP + budget/date verification; lead routing with 5-vendor cap; vendor Android app (lead inbox, calendar, responses); booking-gated reviews; admin console & moderation; programmatic SEO pages + sitemaps; WhatsApp/SMS/push notifications; core analytics	Photography wedge features (style tags, availability, package cards, portfolio layout) ride existing epics — no new headcount
Should — by Jul 2027	Escrow payments & booking flow (venues first, photographers second); quotes; subscription tiers & GST billing; vendor performance dashboard; e-invites v1; planning checklist; real-wedding stories (photographer-contributed, all-vendor tagging)	Stories seed from the anchor studio's 50 season weddings; gated on funnel health
Could — H2 FY28	Corporate RFP pilot; featured placements / ads beta; referral credits; Hindi UI; richer planning tools	Each gated on a metric (ads beta needs 10k monthly enquiries)
Won't — first year	iOS app; AI recommendations; in-app chat/calls; multi-country; unverified self-serve vendor signup	Revisit at Series A

3. User types & permissions

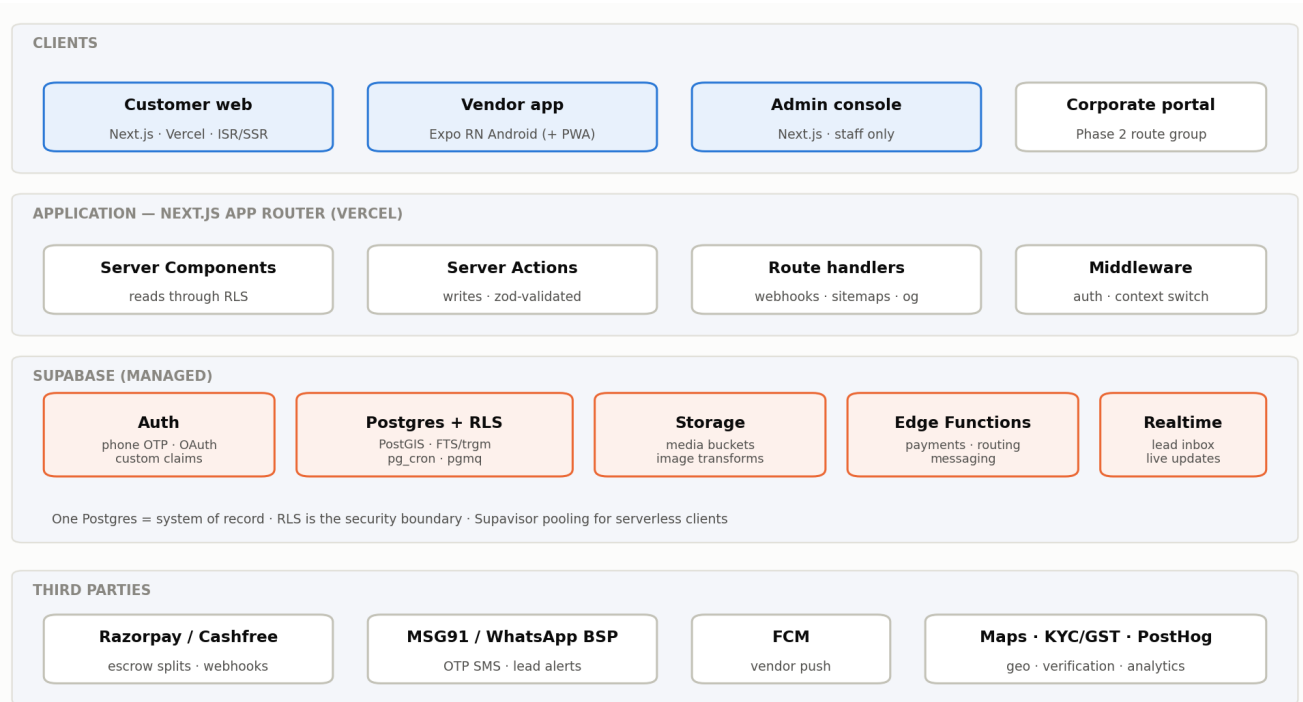
One human = one auth identity; capabilities come from **memberships**, not a role column. A photographer booking a caterer for his own anniversary is one login in two contexts. Model: auth.users 1:1 profiles, plus vendor_members (owner / manager / responder), corporate_members (admin / requester, Phase 2) and staff_roles (field agent, moderator, finance, super) carried in JWT claims. The UI is a context switcher; the database enforces everything.

Type	Who / how created	Primary surface
Guest	Anonymous browsing; the SEO audience	Customer web — discover, profiles, galleries
Customer	Guest becomes Customer on first OTP-verified enquiry	Customer web — events, shortlists, bookings, reviews
Vendor — Owner	Business principal; claims field-created profile via OTP; owns billing, payouts, KYC, team	Vendor app / partner PWA — everything
Vendor — Manager	Invited by owner; edits profile, packages; responds to leads	Vendor app — all except billing/payouts/team
Vendor — Responder	Invited staff; lead inbox and calendar only	Vendor app
Corporate Admin / Requester	Phase 2; Google OAuth with domain capture	Corporate portal — RFPs, panels, billing
Staff (4 roles)	Field agent · moderator/support · finance (four-eyes) · super admin; email + MFA	Admin console — separate deploy, SSO + IP allowlist, append-only audit log

Capability	Guest	Cust	V-Own	V-Mgr	V-Rsp
Browse, search, view profiles	✓	✓	✓	✓	✓
Create enquiry (OTP-gated)	→ Cust	✓	✓	✓	✓
Events, shortlists, checklist	—	✓	✓	✓	✓
See routed leads; respond, quote	—	—	✓	✓	✓
Edit profile, packages, media	—	—	✓	✓	—
Team, subscription, payouts, KYC	—	—	✓	—	—
Write review (completed booking)	—	✓	✓	✓	✓

Corporate roles (Phase 2) add RFP create/publish and panel/billing management for admins, draft-only for requesters.

4. System architecture — Next.js + Supabase



Vercel hosts the Next.js surfaces; Supabase is the entire backend. RLS — not an API layer — is the security boundary between clients and data.

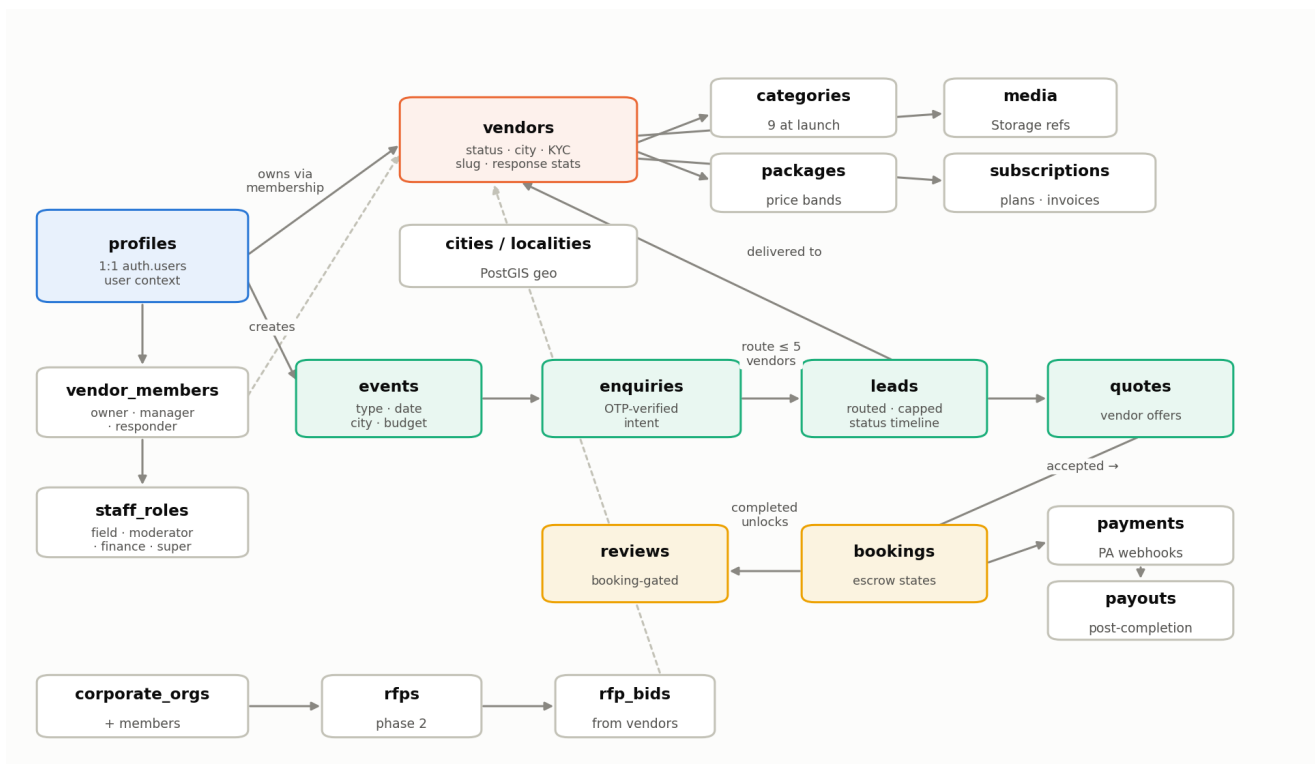
Key decisions: **TypeScript everywhere** (Next.js web, server actions as the API, Expo vendor app) — one hiring pool, shared types end-to-end. **SEO is architecture**: server-rendered ISR pages generated from the catalog with quality thresholds; listing edits revalidate by tag. **Escrow rides a licensed payment aggregator** (Razorpay Route / Cashfree Easy Split — evaluate at build); Utsava never holds customer money. **Postgres-native search first** (FTS + pg_trgm + PostGIS ranking function); Typesense only past mechanical exit criteria (p95 > 200 ms or 50k listings). **Vendor app is Android-first Expo**, shipped after the partner PWA proves the workflows. Platform bill ≈ US\$150–400/month pre-launch (Supabase Pro ~US\$25 base + compute; Vercel; messaging) — well inside the ₹15 L cloud line.

4.1 Repository shape

```
utsava/  
├─ apps/web          # customer web + partner PWA (Next.js 15 App Router)  
│   └─ app/(marketing) · (discover)/[city]/[category]/[locality] · vendor/[slug]  
│       (account) · partner/ · corp/ · api/ (webhooks, sitemaps, og)  
├─ apps/admin        # staff console – separate deploy, SSO + IP allowlist  
├─ apps/vendor-app    # Expo React Native, Android-first  
├─ packages/db        # generated Supabase types, zod schemas, query helpers  
├─ packages/ui        # design system (Tailwind + shadcn/ui)  
└─ supabase/          # migrations (SQL, PR-reviewed) · functions/ · seed/ · pgTAP tests
```

Conventions: Server Components do all reads (RLS-scoped); each feature owns one zod-validated actions.ts; client-side Supabase only for Realtime and Storage uploads; money paths only in Edge Functions / route handlers with signature verification.

5. Data model



The demand spine (green) is the product: events → enquiries → leads → quotes → bookings. Supply (orange) hangs off vendors; money and trust (yellow) hang off bookings; corporate (Phase 2) ends in the same vendor graph.

Domain	Tables (~36 total)
Identity	profiles, staff_roles, audit_log (append-only)
Geo	cities, localities (PostGIS; locality = the SEO unit)
Catalog	categories, vendors, vendor_members, vendor_categories, packages, media, vendor_faqs — style tags live on vendor_categories/packages
Demand	events, enquiries, leads (UNIQUE enquiry × vendor), lead_activities, shortlists
Transaction	quotes, bookings, payments, payouts, refunds, disputes — service-role writes only; money in integer paise
Trust	reviews (FK booking_id UNIQUE — one per completed booking), review_media, moderation_queue, stories (real-wedding galleries, all-vendor tagging)
Monetisation	plans, subscriptions, subscription_invoices (GST-ready)
Corporate (Ph 2)	corporate_orgs, corporate_members, rfps, rfp_invites, rfp_bids
Engagement / Ops	checklists, checklist_items, notifications (outbox), e_invites, onboarding_pipeline, cms_pages

Status enums that drive the product: leads (routed → viewed → responded → quoted → converted / expired), bookings (initiated → advance_held → confirmed → completed → released, with refund_pending / refunded / disputed branches), vendors (draft → pending_review → live → paused / suspended), subscriptions (trial → active → past_due → cancelled).

6. Security: RLS design & compliance

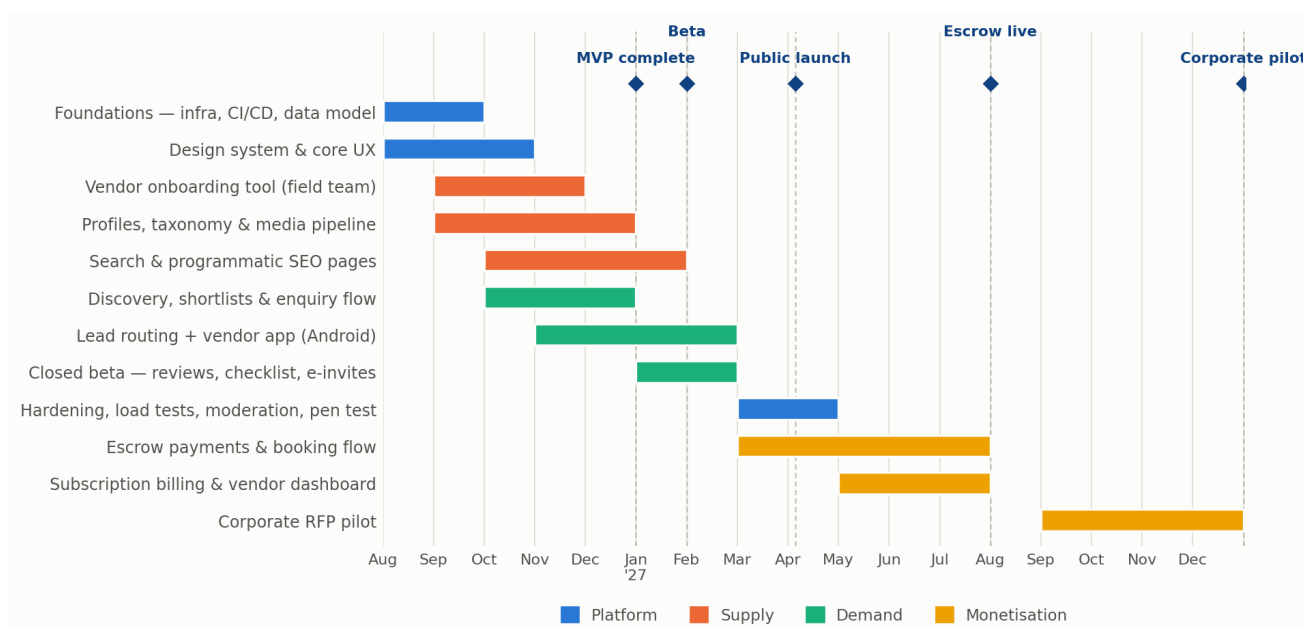
RLS is the authorization model — there is no trusted API between clients and Postgres for reads. Golden rules: **default-deny** on every table; **money tables never trust the client** (payments, payouts, refunds, subscriptions written only by service-role code, merely readable by their parties);

contact masking is a view — vendors see a customer's phone only in permitted lead states;
policies are tested code — a pgTAP suite asserts who sees what, in CI on every migration.
 Cross-table writes (route a lead, accept a quote, transition escrow) are SECURITY DEFINER functions or service-role server actions; the service-role key never ships in a client bundle.

Table(s)	Public / customer	Vendor member	Staff
vendors, packages, media, stories	SELECT where live/published	ALL on own vendor via membership (role-gated)	field: draft-only, own city; moderator: approve
events, enquiries, shortlists	own rows ALL	—	SELECT
leads (via view)	own via enquiry	SELECT own vendor's; contact state-gated; transitions via RPC	R/W
quotes	SELECT own	ALL own vendor's	SELECT
bookings + money tables	SELECT own	SELECT own vendor's	finance via service-role console
reviews	INSERT if own booking completed; SELECT published	SELECT own vendor's	moderator R/W
subscriptions, invoices	—	owner SELECT own	finance R/W

Compliance is built in, not bolted on: DPDP consent at enquiry with purpose limitation (a lead shared with five vendors says so), PII encrypted at rest, deletion SLA honoured, data map maintained; vendor KYC (PAN/GST) gates payouts; escrow rides licensed payment aggregators so PCI scope stays external; an external penetration test runs March 2027 as a launch blocker.

7. Delivery plan & timeline



Workstreams Aug 2026 – Dec 2027. Supply tooling (orange) precedes demand product (green); monetisation (yellow) follows launch. Diamonds are the fixed milestones.

7.1 Cadence, capacity and the sprint map

Two-week sprints, trunk-based, demo every close. Five engineers hands-on (tech lead + four full-stack; mobile joins S6) give ~14 dev-weeks per sprint; the Must-tier backlog totals ~140 dev-weeks — exactly ten sprints from late August to MVP-complete mid-December. Zero schedule slack, so **scope is the release valve**: e-invites and checklist fall back to the beta window before anything touching lead quality. Post-launch, 40% of capacity is reserved for operations and iteration; escrow runs on the rest.

Sprint	Dates	Goal (demoable outcome)
S0	Aug 10 – 21	Team assembled; repos, CI/CD, Supabase envs, design tokens; data-model workshop locked in SQL
S1	Aug 24 – Sep 4	Auth + profiles + RLS harness (pgTAP in CI); admin skeleton; design system v1
S2	Sep 7 – 18	Vendor/listing entities; category taxonomy incl. photography style tags; media pipeline (Storage + transforms)
S3	Sep 21 – Oct 2	Vendor onboarding tool v1 in field hands — portfolio-first profile editor, photo capture, price bands
S4	Oct 5 – 16	Search v1 (FTS + ranking fn); public profiles; SEO page engine v1 + sitemaps; package cards (per-day pricing)
S5	Oct 19 – 30	Discovery UX (filters incl. 'free on my date'); shortlists; onboarding tool v2 (bulk media, offline-tolerant)
S6	Nov 2 – 13	Enquiry flow: OTP + budget/date capture; lead object & states; first end-to-end funnel demo; mobile engineer starts Expo shell
S7	Nov 16 – 27	Lead routing engine with 5-vendor cap; WhatsApp/SMS/FCM via outbox + pgmq; vendor app alpha (inbox, Realtime)
S8	Nov 30 – Dec 11	Vendor app beta (calendar, responses, quotes); booking-gated review model; moderation queue
S9	Dec 14 – 24	Analytics events + funnel dashboard; hardening; content tooling — MVP feature-complete
S10	Jan 4 – 15	Beta cohort live (studio = vendor #1); checklist v1; e-invites v1; stories v1 seeded from studio weddings
S11-S12	Jan 18 – Feb 26	Beta iteration on funnel data; stories publishing flow; vendor dashboard v1; 200-customer cohort complete
S13-S14	Mar 1 – 26	Load test at 5× launch traffic; pen-test remediation; moderation staffing; launch rehearsal + rollback drill

Public launch targets the first half of April 2027, ahead of the Apr–Jun social season. Escrow (venues first, photographers second) lands S15–S21, July 2027.

7.2 Must-tier backlog (≈140 dev-weeks)

Epic	Wks	Epic	Wks
Catalog: profiles, taxonomy, media	16	Vendor app (Expo, Android)	16
Search, ranking & SEO engine	14	Admin console & moderation	12
Discovery, availability filter, shortlists	13	Enquiry & lead verification	10
Vendor onboarding tool	10	Identity, RLS harness & accounts	10
Lead routing & caps	8	Reviews & trust	8
Design system	8	Notifications (outbox + pgmq)	6
Analytics & instrumentation	6	Package cards & pricing display	3

Planning-grade estimates ($\pm 30\%$); the scope valves absorb variance. Real-wedding stories (~ 6 wks) sit in the beta window, not the Must tier.

8. Team & build budget

Role	Count	₹ L / yr	Notes
Tech lead / architect	1	36	Hands-on; owns RLS design, payments, ranking
Full-stack engineers (TS)	4	15 each	Next.js + Supabase across web, partner PWA, admin
Mobile engineer (Expo)	1	15	Joins S6; Android-first vendor app
Product designer	1	12	Design system, vendor & customer UX
QA engineer	1	9	Automation-first; owns device lab
Product manager	1	16	Founder is head of product above this
DevOps (fractional)	0.3	6 (contract)	Vercel/Supabase config, observability, on-call setup

Annual run-rate $\approx$ ₹1.54 Cr all-in (seed-stage, ESOP-weighted). With hiring staggered across August–September 2026 and mobile joining in November, 18-month people cost $\approx$ ₹2.2 Cr. The remainder of the ₹2.8 Cr: cloud & third-party services ₹12 L (managed stack), tools & licences ₹8 L, device lab ₹5 L, contingency ₹35 L — a fatter buffer than v1 precisely because the managed stack removed infra spend. Hiring sequence: tech lead + one full-stack sign before seed close; designer no later than S1.

9. Engineering practices, QA & environments

Quality. Unit tests on domain logic (lead routing, caps, escrow state machine at $\sim 90\%$ — money and trust paths are non-negotiable); pgTAP RLS suite per table; Playwright end-to-end on the ten critical journeys on every merge; vendor app tested on a physical low-end-Android device lab; beta UAT scripted through the field team. **Release.** Trunk-based with feature flags; weekly release train post-launch with canary + rehearsed rollback; on-call starts with the March hardening phase.

Environments. Local supabase start + seed; per-PR Vercel preview paired with a Supabase preview branch; separate staging and production projects; production schema changes only via CI-applied, PR-reviewed SQL migrations; CI gates on generated-type drift, pgTAP, E2E and Lighthouse budgets; nightly logical backups exported externally on top of PITR. **Definition of done:** reviewed, tested, events emitting, dashboards updated, flags documented, accessibility pass — and an abuse-case review for anything touching leads, money or reviews.

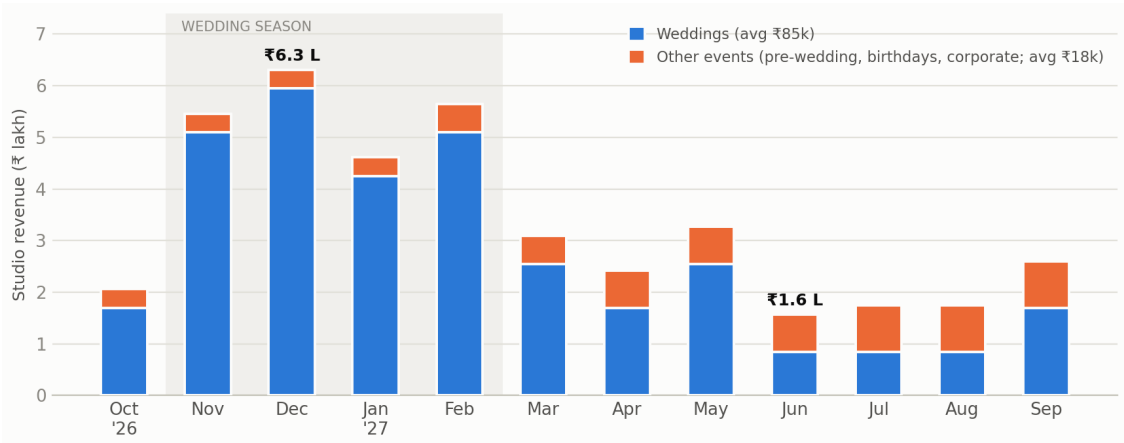
10. Instrumentation & KPIs

Funnel stage	Core events	KPI it feeds
Discover	search_performed, listing_viewed, shortlist_added	10k monthly enquiries gate; SEO yield; photography ≥ 40% of enquiries
Enquire	enquiry_submitted, otp_verified, lead_qualified	Verification rate; cost per verified lead
Match	lead_routed, vendor_viewed_lead, vendor_responded	Cap compliance; median vendor response < 2 h
Transact	quote_sent, booking_initiated, escrow_paid, booking_completed	Lead→booking ≥ 5% (photography ≥ 6%); managed GMV
Trust	review_requested, review_published, story_published, dispute_opened	Review coverage; stories/month (content flywheel); disputes < 1%
Monetise	subscription_started, subscription_renewed, invoice_paid	180 paying vendors by Mar 2028; renewal ≥ 55-60%

North star: **verified leads delivered per week**. Events flow to PostHog + a nightly warehouse export; vendor dashboards and board dashboards read the same semantic definitions, so the numbers always match.

11. The photography wedge in this plan

Strategy lives in the Photography-First Strategy document; here is what it changes in the build. **Scope:** style taxonomy (S2), portfolio-first profiles (S3), package cards (S4), availability filter (S5) — absorbed into existing epics — plus real-wedding stories in the beta window. **Supply targets:** 800 of 3,000 launch listings are photographers with complete portfolios and price bands. **The anchor studio** (founder's own service, separate brand and P&L;) operates from October 2026: it funds the build season (~₹40 L year-one revenue), produces the 50 real-wedding stories the platform launches with, and serves as beta vendor #1 — the first user of leads, quotes, calendar, reviews and, in July 2027, escrow end-to-end. On-platform it runs under disclosed, overflow-first rules with no ranking favour (auditable in the ranking SQL). Guardrail: the founder shoots at most one wedding a week, a studio manager runs day-to-day from November, and platform milestones win every clash.



Anchor studio year one (Oct 2026 – Sep 2027): ≈ ₹40.5 L across 39 weddings + 41 event shoots; the off-season survives on the all-events layer — the platform thesis in miniature.

12. Consolidated risks & mitigations

Risk	Mitigation
RLS policy mistakes — a wrong policy is a silent data leak	Default-deny; pgTAP per table in CI; security review each phase gate; contact fields only via views/RPC
SEO indexing lags launch — demand engine starts cold	Page engine live from Oct 2026 so indexing starts pre-launch; quality thresholds; sitemap partitioning
Lead-quality failure — kills renewal, the model's canary	Verified leads, 5-vendor cap, credit refunds, honest dashboards; renewal gates every phase
Founder bandwidth — studio in season + platform in build	Studio manager by Nov 2026; founder ≤ 1 shoot/week; platform wins every clash
Channel-conflict backlash — vendors discover the platform runs a studio	Disclosure badge, same ranking SQL, overflow-first posture, ~5% category-booking cap, published policy
Escrow edge cases — stuck refunds at someone's wedding	Explicit state machine + timeouts; PA sandbox soak tests; dispute runbook; audited manual override
Connection exhaustion / serverless limits	Supervisor transaction pooling everywhere; no long-lived connections in functions
Postgres search ceiling	Mechanical exit criteria to Typesense; ranking isolated behind one SQL function
Vercel/media cost creep at SEO scale	Supabase CDN transforms bypass the image optimizer; static-first pages; budget alarms
Key-person dependency in a 5-engineer core	Docs-as-you-go; module ownership pairs; hiring pipeline warm

13. Launch readiness — go/no-go, late March 2027

- **Supply:** 3,000 live listings (≥ 800 photographers) with ≥ 5 photos and price bands; 500 beta-verified reviews; 50 real-wedding stories; every category ≥ 50 listings per city.
- **Funnel:** search $\rightarrow$ verified lead $\rightarrow$ vendor response at $\geq 99\%$ technical success in beta; median vendor response under two hours.
- **Performance:** 5 $\times$ load test passed; LCP < 2.5 s on 4G mid-range Android; API p95 < 300 ms, search < 200 ms.
- **Safety:** pen-test findings remediated; DPDP consent flows reviewed; moderation staffed with SLA; RLS suite green.
- **Operations:** on-call live; rollback rehearsed; support runbook published; dashboards reviewed weekly with founders.

14. Post-launch roadmap & milestones

Window	Focus	Gate it serves
Apr – Jul 2027	Escrow bookings (venues $\rightarrow$ photographers); quotes; subscription billing + GST; vendor dashboard	First paid subscriptions Jul 2027; commission revenue starts
Aug – Dec 2027	Corporate RFP pilot; stories flywheel; referral credits; Hindi groundwork; second crew demand for studio	20 corporate accounts by Dec 2027; 10k monthly enquiries
Jan – Mar 2028	Ads/featured beta; search relevance v2 (behavioural); cost & performance tuning	FY28 close: ₹1.3 Cr revenue, 180 paying vendors, NPS > 50

Window	Focus	Gate it serves
FY29	iOS app; 8-city scale-out (config-driven launches); data platform (warehouse, ML ranking); module extractions if load demands	Series A metrics; ₹6.35 Cr revenue year

Document map: **Business Plan v1** (market, competition, financial model) · **this document** (the complete build reference) · **Photography-First Strategy** (category playbook + studio growth engine). All dates assume seed close by September 2026; a slip shifts the timeline one-for-one but never the sequence. Suggested next artifact: the initial Supabase migration pack — schema, enums, RLS policies and seed data, ready for supabase migration new.

Key sources carried from the companion documents: Grand View Research & IBEF (market), Inc42 (WedMeGood financials), Mordor Intelligence (events market), VelvetKnot/VenueLook/Ollar (photography pricing 2026), Navdeep Soni (vendor-side portal economics), makerkit.dev & nocode.mba (Supabase pricing 2026).